STATS102A Midterm Study Guide

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Chapter 1

- Basic Data Structure
  - Atomic vectors
  - Generic vectors
- Coercion
  - Explicit coercion
  - Implicit coercion
- Special Values: NA, NULL, and NaN
- Vector Recycling
- Subsetting
- Control Statement
- Functions
  - Prefix
  - Infix
  - Replacement
  - Special
- Error handling

Chapter 2

- Scoping
  - Name masking
  - A fresh start
  - Dynamic lookup
- Environment objects
  - All objects created and stored in an R session are collectively called the workspace.
  - The body of a function creates a local environment.
  - The parent is used to implement lexical scoping.
- Package environment
- Namespace environment
- Function environment
- Execution environment
- Calling environment

Chapter 3

- Creating, printing, and subsetting Tibbles
- The readr package
- Tidy data:
  - Each variable must have its own column.
  - Each observation must have its own row.
  - Each value must have its own cell.

  There are four functions that address common issues with messy data:
  - pivot_longer: When one variable is spread across multiple columns.
  - pivot_wider: When one observation is scattered across multiple rows.
  - separate: When cells contain multiple values (from different variables).
  - unite: When a single variable is spread across multiple columns.

Chapter 3(Cont.)

- Data Manipulation with dplyr
  - filter() extracts rows (observations) by their values.
  - select() extracts columns (variables) by their names.
  - arrange() reorders rows.
  - summarize() makes summarizing by groups easier.
  - mutate() creates new variables from existing variables.
- Relational Data
  - A primary key uniquely identifies an observation in its own table.
  - A foreign key uniquely identifies an observation in another table.
  - A primary key and the corresponding foreign key in another table form a relation.

  To work with relational data you need verbs that work with pairs of tables:
  - Mutating joins: inner_join() and outer_join()
  - Filtering joins: semi_join() and anti_join()
- Functional Programming with purrr: The **map** family of functions that apply a function to each element of a vector.